

Measuring Inter-Annotator Agreement

Can You Trust Your Gold Standard?

November 29th, 2018

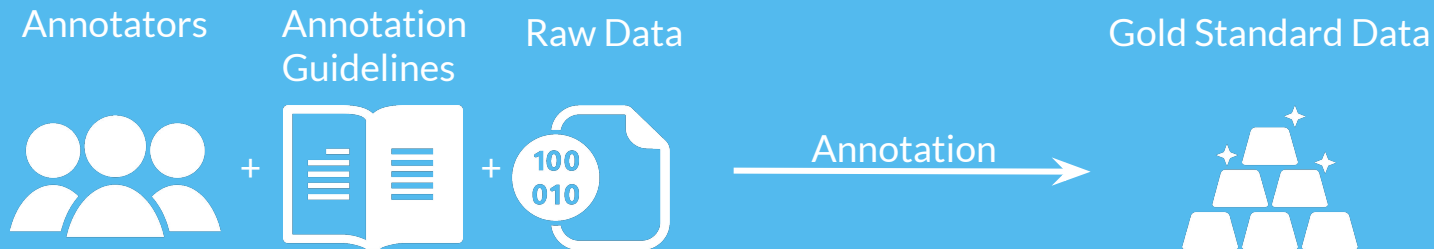
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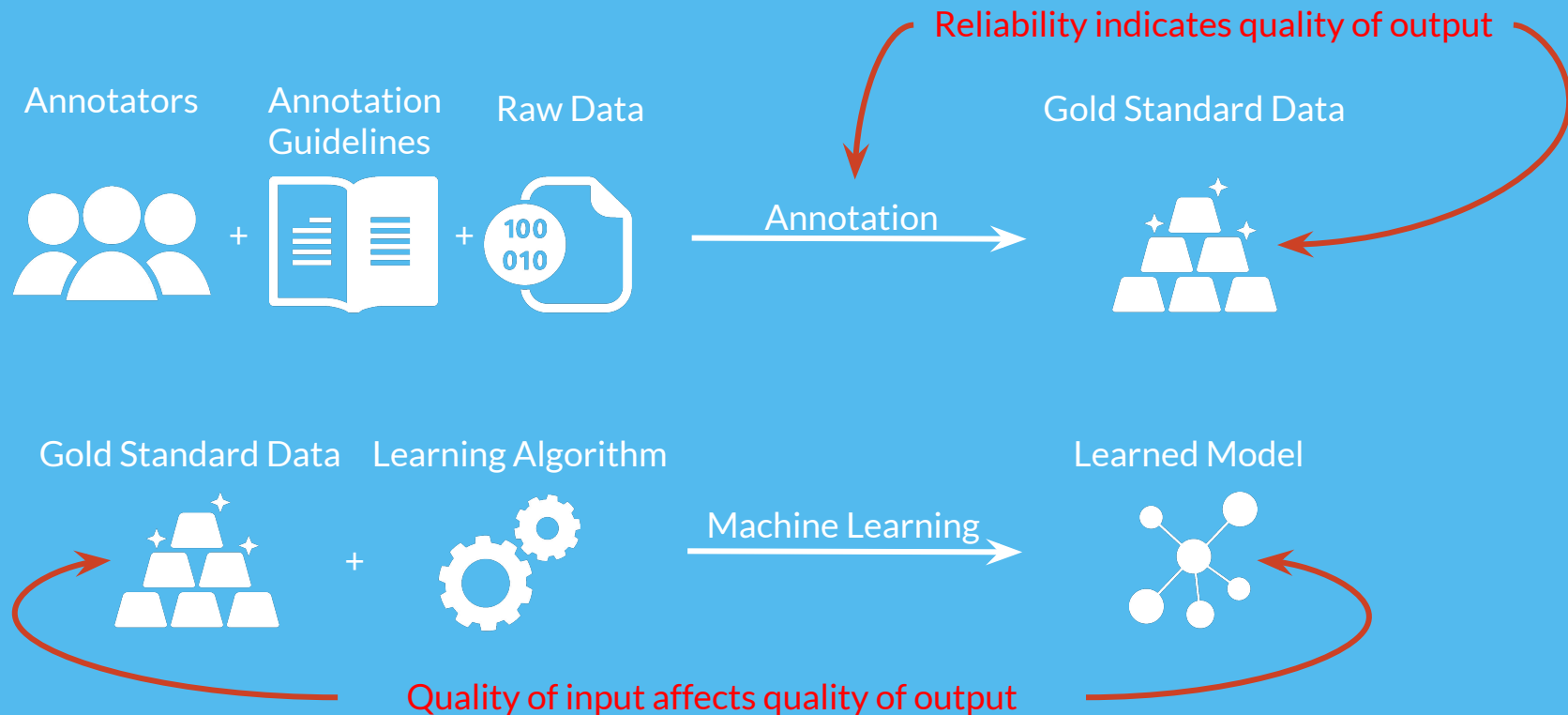


BASIS
TECHNOLOGY

Blueprint for a Machine Learning Application



Blueprint for a Machine Learning Application





Pyrite



Gold

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Some IAA Metrics

1. *F* Measure
2. Kappa (κ)
3. Alpha (α)



F Measure

One option is to use the same evaluation metric often used to evaluate machine generated annotations against a gold-standard.

F Measure is defined as:

$$F_{\beta} = (1 + \beta^2) \frac{Precision \cdot Recall}{(\beta^2 \cdot Precision) + Recall}$$

$$Precision = \frac{|true\ positives|}{|true\ positives| + |false\ positives|}$$

$$Recall = \frac{|true\ positives|}{|true\ positives| + |false\ negatives|}$$

F Measure

	Annotator		Error Type	
Example Data	A	B	Gold=A/Prediction=B	Gold=B/Prediction=A
How can I be productive when I can't sleep til 6.30 am and I'm a zombie the next day :)			 : False positive  : False negative	 : False positive  : False negative
bethhh i have missed u sm todayyy!! :(thank u thank u ur so cute			 : False positive  : False negative	 : False positive  : False negative

F Measure

- The range of F scores is **[0.0 ... 1.0]** with higher values indicating higher accuracy and **1.0** indicating **perfect accuracy**.
- **Pro:** F measure is sensitive to both false negative and false positive error types.
- **Con:** Measures accuracy with respect to a gold standard, which assumes a gold standard exists (care is required when interpreting the results!)
- **Con:** Cannot be easily compared across different annotators/tasks/datasets

Kappa

- Cohen (1960) introduced an IAA metric with a desirable property. For a pair of independent, parallel annotations, kappa takes into account:
 - p_o = **observed** agreement
 - p_e = **expected** agreement
- Fleiss (1971) introduced a generalized form of kappa which allows for measuring agreement across a set of two or more annotators.

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

What if these guys are your ~~annotators~~ villains?



Anton Chigurh



Guino Rinaldo



Harvey Dent

Kappa

- The range of kappa scores is **[-1.0 ... 1.0]** with higher values indicating higher reliability.
- **1.0** indicates perfect agreement
- **0.0** indicates agreement no better than random chance
- **Pro**: Measures the level of agreement above chance, thus is comparable across tasks, annotators, or datasets.
- **Pro**: Measures IAA between 2 or more annotators.
- **Con**: Only captures agreement for nominal labels.

Alpha

- Krippendorff (1970) introduced another IAA metric with several desirable properties. For any number of parallel annotations, kappa takes into account:
 - D_o = **observed** agreement
 - D_e = **expected** agreement

$$\alpha = 1 - \frac{D_o}{D_e}$$

Alpha

- Alpha also takes, as a parameter, a delta function that specifies a distance metric between two annotations.
- I.e., you can define and supply any distance metric of the form:

$$\delta = \lambda(a, b) \rightarrow [0.0 \dots \infty]$$

Alpha

- One can define any number of delta metrics, e.g.:

$$\delta_{nominal}(a, b) = \begin{cases} 0 & \text{if } a = b \\ 1 & \text{if } a \neq b \end{cases}$$

$$\delta_{ordinal}(a, b) = \left(\sum_{i=a}^{i=b} n_i - \frac{n_a + n_b}{2} \right)^2$$

Alpha

- A nominal scale treats any two different data points as equally different:

A	B	$\delta_{nominal}$	$\delta_{ordinal}$
		0	0
		0	0
		0	0

A	B	$\delta_{nominal}$	$\delta_{ordinal}$
		1	1
		1	2
		1	1

Alpha

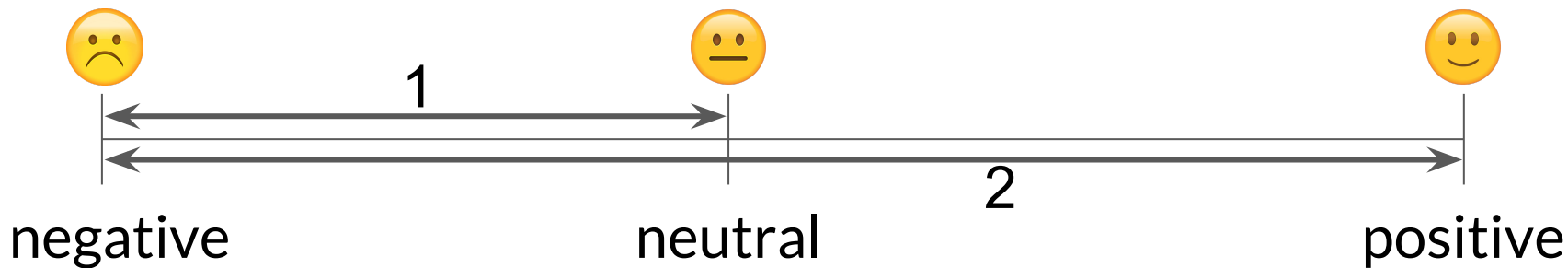
- A nominal scale treats any two different data points as equally different:

A	B	$\delta_{nominal}$	$\delta_{ordinal}$
		0	0
		0	0
		0	0

A	B	$\delta_{nominal}$	$\delta_{ordinal}$
		1	1
		1	2
		1	1

Alpha

- An ordinal scale accounts for the relative difference in values based on their order:



Alpha

$$\delta_{\text{ordinal}}(\star\star, \star\star\star\star\star) = 3$$

$$\delta_{\text{Jaccard}}(\{\text{politics}, \text{sports}\}, \{\text{sports}\}) = 0.33$$

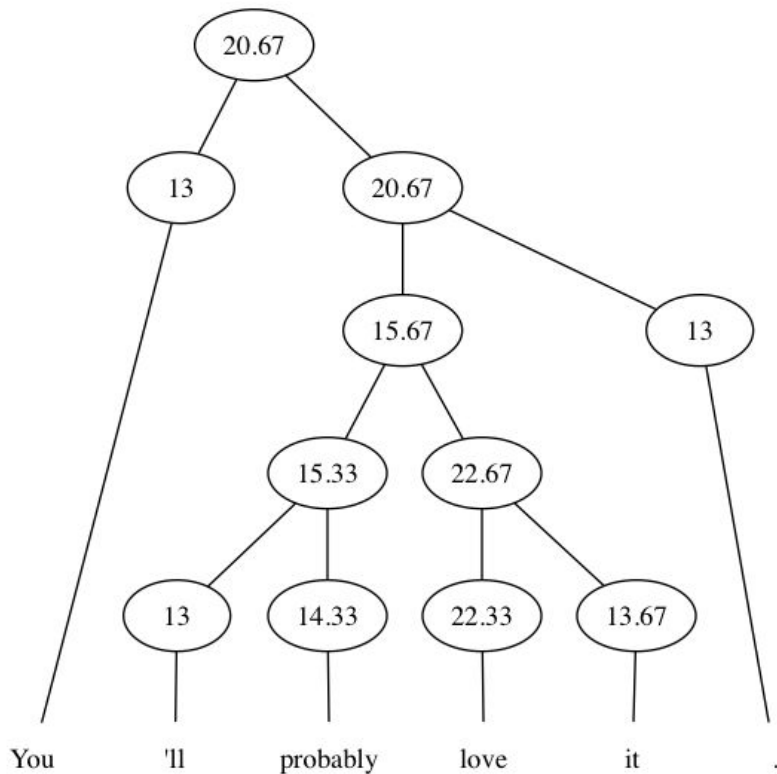
$$\delta_{\text{time}}(12:51 \text{ pm}, 1:51 \text{ pm}) = 60$$

$$\delta_{\text{lat/long}}((40.7, -74.0), (51.5, -0.13)) = 3470$$

$$\delta_{\text{Euclidean}}(\text{\#00A2FF}, \text{\#FFA200}) = 360.62$$

$$\delta_{\text{Levenshtein}}(\text{"New York City"}, \text{"New York"}) = 5$$

Alpha



- The **Stanford Sentiment Treebank**, (Socher et. al. 2013) consists of sentiment ratings applied to nodes in syntactic parse trees on an **ordinal scale** from [1 ... 25]:

Distance metric	α Score
$\delta_{nominal}$	0.19
$\delta_{ordinal}$	0.57

Alpha

- The range of alpha scores is **[-1.0 ... 1.0]** with higher values indicating higher reliability.
- **1.0** indicates perfect agreement
- **0.0** indicates agreement no better than random chance
- **Pro**: Measures the level of agreement above chance, thus is comparable across tasks, annotators, or datasets.
- **Pro**: Measures IAA between 2 or more annotators.
- **Pro**: Can account for any data type.
- **Pro**: Is tolerant of missing data (in case not all annotators annotated every data point)

Lombard et. al. (2004):

*“Coefficients of **0.90** or greater are **nearly always acceptable**, **0.80** or greater is **acceptable in most situations**, and **0.70** may be appropriate in some exploratory studies for some indices.”*

*“A preferred approach is to calculate and **report two (or more) indices** ...”*

Krippendorff (2004):

*“... to be sure that the data under consideration are at least similarly interpretable by other scholars ... require $\alpha \geq 0.80$, and where **tentative conclusions** are still acceptable, $\alpha \geq 0.667$...”*

Interpreting Scores

- If we can't achieve reliable annotations, we can't expect our dataset to be useful for evaluation (or in the case of supervised machine learning, for training).
- If annotations are unreliable, it may indicate that the annotation task is not well defined, the annotation guidelines were unclear, or that our annotators were not capable of performing the task.

Review of IAA Metrics

Metric	Range	Data Type	# Annotators	Accounts for Chance Agreement
F_{β}	[0.0 ... +1.0]	nominal*	2 (pair-wise)	no
κ	[-1.0 ... +1.0]	nominal	2+	yes
α	[-1.0 ... +1.0]	any	2+	yes

* F_{β} can be modified to allow partial credit dependent on error type which can account for data types beyond nominal values

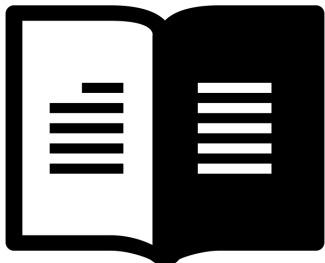
Takeaways

- When creating your own dataset, *always* measure IAA!
- When assessing public datasets, *confirm* published IAA scores!
- Be wary of any dataset which does not publish reliability scores! **Don't rely on unreliable data!**
- Interpret measurements carefully; think about what you're measuring and why!

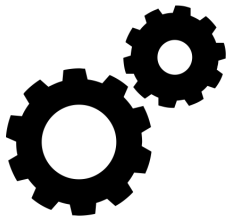
Sources

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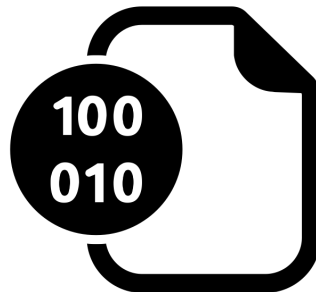
The Noun Project



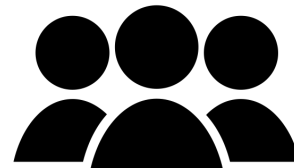
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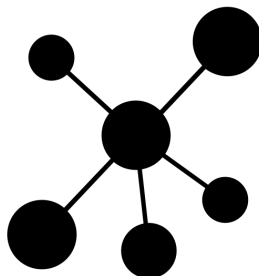
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Thank you!

Questions?

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